

Missed

S3 1.a. $\sum_{n \geq 1} \frac{2}{3^n} = \frac{2}{3} + \frac{2}{3^2} + \frac{2}{3^3} + \frac{2}{3^4} + \dots = \frac{2}{3} \left(1 + \frac{1}{3} + \frac{1}{3^2} + \dots \right) \because q = \frac{1}{3} < 1.$
 $= \frac{2}{3} \cdot \frac{1}{1 - \frac{1}{3}} = \frac{2}{3} \cdot \frac{1}{\frac{2}{3}} = 1.$

1.d. $\sum_{n \geq 1} \frac{1}{n(n+1)(n+2)} = \frac{1}{2} \sum_{n \geq 1} \frac{n+2-n}{n(n+1)(n+2)} = \frac{1}{2} \sum_{n \geq 1} \left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right)$
 $\lim_N \frac{1}{2} \cdot \left(\frac{1}{1 \cdot 2} - \frac{1}{2 \cdot 3} + \frac{1}{2 \cdot 3} - \frac{1}{3 \cdot 4} + \dots + \frac{1}{N(N+1)} - \frac{1}{(N+1)(N+2)} \right)$
 $= \lim_N \frac{1}{2} \left(\frac{1}{1 \cdot 2} - \frac{1}{(N+1)(N+2)} \right) = \frac{1}{2} \cdot \left(\frac{1}{2} - 0 \right) = \frac{1}{4}.$

1.e. $\sum_{n \geq 1} \frac{n}{2^n} = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots$
 $= \frac{1}{2} + \frac{1}{2} \cdot \left(\frac{2}{2} + \frac{3}{2^2} + \dots + \frac{n}{2^{n-1}} + \dots \right)$
 $= \frac{1}{2} + \frac{1}{2} \cdot \left(\frac{2-1+1}{2^{2-1}} + \frac{3-1+1}{2^{3-1}} + \dots + \frac{n-1+1}{2^{n-1}} + \dots \right)$
 $= \frac{1}{2} + \frac{1}{2} \cdot \left(\frac{1}{2} + \frac{1}{2} + \frac{2}{2^2} + \frac{1}{2^2} + \dots + \frac{n-1}{2^{n-1}} + \frac{1}{2^{n-1}} + \dots \right)$
 $= \frac{1}{2} + \frac{1}{2} \cdot \left[\underbrace{\left(\frac{1}{2} + \frac{2}{2^2} + \dots + \frac{n-1}{2^{n-1}} \right)}_S + \underbrace{\left(\frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}} \right)}_1 \right]$
 $\Rightarrow S = \frac{1}{2} + \frac{1}{2} \cdot (S+1) \Rightarrow S = \frac{1}{2} + \frac{1}{2}S + \frac{1}{2} \Rightarrow \frac{1}{2}S = 1 \Rightarrow \underline{\underline{S=2}}.$

$\frac{1}{2} + \frac{1}{2^2} + \dots = \frac{1}{2} \left(1 + \frac{1}{2} + \dots \right) = \frac{1}{2} \cdot \frac{1}{1 - \frac{1}{2}} = \frac{1}{2} \cdot \frac{1}{\frac{1}{2}} = 1.$

1.f. $\sum_{n \geq 1} \frac{1}{3\sqrt{n}} = \sum_{n \geq 1} \frac{1}{n^{\frac{1}{2}}} < 1 \Rightarrow$ divergent.

2.a. $\sum_{n \geq 2} \frac{1}{\ln n}$ "like" $\sum_{n \geq 2} \frac{1}{n}$ div.

$\lim_n \frac{\left(\frac{1}{\ln n} \right)^{x_n}}{\left(\frac{1}{n} \right)^{y_n}} = \lim_n \frac{n}{\ln n} = \infty \Rightarrow$ if y_n diverges $\Rightarrow x_n$ div.

2.c. $\sum_{n \geq 1} \frac{\ln(1 + \frac{1}{n})}{n}$ like $\sum \frac{1}{n^2} \xrightarrow{>1} \text{conv.}$

$$\lim_n \frac{\frac{\ln(1 + \frac{1}{n})}{n}}{\frac{1}{n^2}} = \lim_n \frac{\ln(1 + \frac{1}{n})}{\frac{1}{n}} \cdot n^2 = \lim_n n \ln(1 + \frac{1}{n})$$

$$= \lim_n \ln\left(1 + \frac{1}{n}\right)^n = \ln \lim_n \left(1 + \frac{1}{n}\right)^n = \ln e = 1. \Rightarrow \text{same nature.}$$

2f. $\sum_{n \geq 2} \frac{1}{n \ln n}$

Cauchy cond.

$$\sum \frac{2^x}{2^x \cdot \ln 2^m}$$

$$= \sum \frac{1}{\ln 2^m}$$

$$= \sum \frac{1}{m \ln 2}$$

$$= \frac{1}{\ln 2} \sum \frac{1}{m}$$

div.

Integral test.

"like" $\int_2^{\infty} \frac{1}{x \ln x} dx$

$$= \lim_{t \rightarrow \infty} \int_2^t \frac{1}{x \ln x} dx = \lim_{t \rightarrow \infty} (\ln(\ln t) - \ln(\ln 2)) = \infty - \dots = \infty$$

$$\ln x = u \Rightarrow \frac{1}{x} dx = du$$

$$x = 2 \Rightarrow u = \ln 2$$

$$x = t \Rightarrow u = \ln t$$

$$\int_{\ln 2}^{\ln t} \frac{1}{u} du = \ln|u| \Big|_{\ln 2}^{\ln t} = \ln(\ln t) - \ln(\ln 2)$$

3.a. $\sum_{n \geq 1} \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot \dots \cdot 2n}$

Raabe. $\lim_n n \cdot \left(\frac{2n+2}{2n+1} - 1 \right) = \lim_n n \left(\frac{2n+2-2n-1}{2n+1} \right) = \lim_n \frac{n}{2n+1} = \frac{1}{2} < 1$

$\Rightarrow$ divergent.

3b. $\sum_{n \geq 1} a^{\ln n}, a > 0$

$$\lim_n n \cdot \left(\frac{x_n}{x_{n+1}} - 1 \right) = \lim_n n \cdot \left(a^{\frac{\ln n}{n+1}} - 1 \right) = \lim_n n \left(\frac{a^{\frac{\ln n}{n+1} - 1}}{\frac{\ln n}{n+1}} \cdot \ln \frac{n}{n+1} \right)$$

$$= \lim_n n \cdot \ln \frac{n}{n+1} = \ln a \cdot \lim_n \ln \frac{n}{n+1} \cdot n$$

$$= \ln a \cdot \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1}\right)^n = \dots$$

$$\frac{a^{\ln n}}{a^{\ln(n+1)}} = a^{\ln n} : a^{\ln(n+1)} = a^{\ln n - \ln(n+1)} = a^{\ln \frac{n}{n+1}}$$

3c. $\sum_{n \geq 1} \frac{a^n \cdot n!}{n^n}, a > 0$

$$\frac{x_{n+1}}{x_n} = \frac{a^{n+1} \cdot (n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{a^n \cdot n!} = a \cdot \left(\frac{n}{n+1}\right)^n$$

$$\lim_n \dots = \frac{a}{e} \begin{cases} \frac{a}{e} > 1 \Rightarrow a > e \Rightarrow \text{div.} \\ \frac{a}{e} < 1 \Rightarrow a < e \Rightarrow \text{conv.} \\ \frac{a}{e} = 1 \Rightarrow a = e \Rightarrow \sum \frac{e^n \cdot n!}{n^n} \rightarrow \dots \text{conv.} \end{cases}$$

54. 1.a. $\sum_{n \geq 1} \frac{(-1)^{n+1}}{\sqrt{n(n+1)}}$ conv.? abs. conv.? Leibniz $\rightarrow$ conv.

$$\sum \left| \frac{1}{\sqrt{n(n+1)}} \right| = \sum \frac{1}{\sqrt{n(n+1)}} \text{ "like" } \sum \frac{1}{n} \text{ div. } \Rightarrow \text{not abs. conv.}$$

$$\frac{\frac{1}{\sqrt{n(n+1)}}}{\frac{1}{n}} = \frac{n}{\sqrt{n(n+1)}} = \frac{\sqrt{n}}{\sqrt{n+1}} \rightarrow 1 \in (0, \infty)$$

1b. $\sum_{n \geq 1} (-1)^n \cdot \sin \frac{1}{n}$ $\left. \begin{array}{l} \sin \frac{1}{n} \dots \downarrow \end{array} \right\} \Rightarrow \text{Leibniz test} \Rightarrow \text{conv.}$

$$\sum \left| \sin \frac{1}{n} \right| = \sum \sin \frac{1}{n} \text{ "like" } \sum \frac{1}{n} \text{ div. } \Rightarrow \text{not abs. conv.}$$

$\frac{\sin \frac{1}{n}}{\frac{1}{n}} \rightarrow 1 \in (0, \infty)$

$$3. \sum_{n \geq 1} \frac{x^n}{n} \stackrel{?}{=} -\ln(1-x)$$

by int. $\frac{1}{1-x} = \sum_{n \geq 0} x^n, |x| < 1$

$$\sum_{n \geq 1} (-1)^n \cdot \frac{x^n}{n} \stackrel{?}{=} \ln(1+x)$$

$$\sum t^n = \frac{1}{1-t}, |t| < 1 \quad \int_0^x$$

$$\int_0^x \sum t^n = \int_0^x \frac{1}{1-t} dt$$

$$\sum \frac{t^{n+1}}{n+1} \Big|_0^x = \int_0^x \frac{1}{1-t} dt$$

$$\sum \frac{x^{n+1}}{n+1} = -\ln|t-1| \Big|_0^x$$

$$\sum_{n=0} \frac{x^{n+1}}{n+1} = -\ln|1-x|$$

$$\left(\sum_{n=1} \frac{x^n}{n} = -\ln(1-x) \right) \xrightarrow{x=-x} \sum_{n \geq 1} \frac{(-x)^n}{n} = -\ln(1+x)$$

$$\sum (-1)^n \cdot \frac{x^n}{n} = -\ln(1+x) \cdot (-1)$$

$$\sum (-1)^{n+1} \cdot \frac{x^n}{n} = \ln(1+x), \quad x \in (-1, 1)$$

$$x=1: \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} \dots \rightarrow \ln 2$$

$$x=-1: \sum_{n \geq 1} \frac{(-1)^{n+2}}{n} = -1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} \dots \rightarrow -\ln 2$$

$$4. \sum_{n \geq 0} (-1)^n \cdot \frac{x^{2n+1}}{2n+1} \stackrel{?}{=} x - \frac{x^3}{3} + \frac{x^5}{5} - \dots = \arctan x, \quad x \in [-1, 1]$$

$$\sum t^n = \frac{1}{1-t}$$

↓

$$t = -t$$

$$\sum (-t)^n = \frac{1}{1+t}$$

↓

$$t = t^2$$

$$\sum (-1)^n \cdot t^{2n} = \frac{1}{1+t^2} \Big/ \int_0^x \Rightarrow \dots \sum (-1)^n \cdot \frac{x^{2n+1}}{2n+1} = \operatorname{arctg} x.$$

↗ $x = -1 \dots$
↘ $x = 1 \dots$

Why is $\operatorname{arctg}(-1) = -\frac{\pi}{4} = \dots (-1)$?

$$\text{b. e. } \sum_{n \geq 1} \frac{(x-1)^n}{n^p}, p > 0$$

$$\lim_n \left| \frac{a_{n+1}}{a_n} \right| = \dots 1 = L \Rightarrow R = 1$$

$$\text{5. f. } \sum_{n \geq 1} \frac{(-1)^n \cdot x^n}{\sqrt{n}} \dots$$

S5. 1. $\mathbb{Z}' = ? \quad \{-\infty, \infty\}$
 $\mathbb{Q}' = ? \quad \bar{\mathbb{R}}$

2. $\begin{cases} f \text{ discont. everywhere} \\ |f| \text{ cont. everywhere} \end{cases}$

$$\begin{cases} 1, & x \in \mathbb{Q} \\ -1, & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

$f \text{ cont. in } \underline{x_0} \Leftrightarrow (\forall) (x_n) \rightarrow x_0,$
 $\lim_{x_n \rightarrow x_0} f(x_n) = f(\lim_n x_n) = x_0.$

$\left. \begin{array}{l} x_n \in \mathbb{Q}, \quad x_n \rightarrow x_0, \quad f(x_n) = 1. \\ \bar{x}_n \in \mathbb{R} \setminus \mathbb{Q}, \quad \bar{x}_n \rightarrow x_0, \quad f(\bar{x}_n) = -1 \end{array} \right\} \Rightarrow \neq \text{discont. at } x_0.$

6f. $\lim_{x \rightarrow \infty} x \left[\left(1 + \frac{1}{x} \right)^x - e \right]$

de $\frac{0}{0} ??$

$$\lim_n \frac{\left(1 + \frac{1}{x} \right)^x - e}{\frac{1}{x}} = \lim_n \frac{e^{\ln \left(1 + \frac{1}{x} \right)^x} - e^1}{\frac{1}{x}} \dots$$

$$4. f(x) = \begin{cases} x^2 \cdot \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Δ

$$\lim_{x \rightarrow 0} f(x) \stackrel{?}{=} f(0)$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} x^2 \cdot \sin \frac{1}{x} = \lim_{x \rightarrow 0} \frac{\sin \frac{1}{x}}{\frac{1}{x^2}} = f(0)$$

$$0 \leq |x^2 \sin \frac{1}{x}| \leq |x^2| \cdot \underbrace{|\sin \frac{1}{x}|}_{\leq 1} \leq x^2$$

f cont.



$$|f(x)| \rightarrow 0 \Rightarrow f(x) \rightarrow 0.$$

$$f'(x) = 2x \sin \frac{1}{x} - \cos \frac{1}{x}$$

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{x^2 \sin \frac{1}{x}}{x} = \lim_{x \rightarrow 0} x \cdot \sin \frac{1}{x} \stackrel{\text{o.b.d}}{=} 0.$$

$$\Rightarrow \begin{cases} 2x \sin \frac{1}{x} - \cos \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases} \Rightarrow \exists \lim, \exists f'(0) = 0$$

f' cont

$$x: 0 : \lim_{x \rightarrow 0} f'(x) = f'(0) \Rightarrow f' \text{ discontinuous at } x_0$$

↓
A

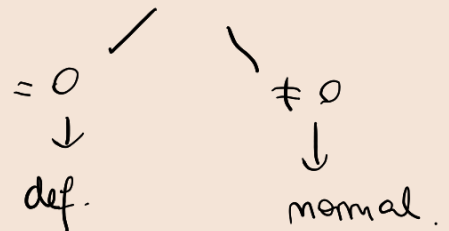
↓
not diff.

f cont? (0)

$$\lim_{x \rightarrow 0} f(x) \stackrel{?}{=} f(0)$$

?? if $\nexists \lim \sin \frac{1}{x}$
then why $\rightarrow 0$?

f diff? (0)



f' cont.

$$\lim_{x \rightarrow 0} f'(x) \stackrel{?}{=} f'(0)$$

f' diff.

S7.

$$1. c. \lim_n \frac{\sqrt[n]{n!}}{n} \ln x_n \dots$$

$$2. f: [0,1] \rightarrow \mathbb{R}, f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases} \quad \text{i.r.? (2 partitions } \rightarrow \text{ same sum.)}$$

$$\text{Partition of } [0,1]: \underline{x_0 = 0} < x_1 < x_2 < \dots < \underline{x_n = 1}$$

$$\text{let } x_k = \frac{k}{n}$$

$$\underline{x_{k-1}} - \underline{x_k} = \frac{2}{n} - \frac{1}{n} = \frac{1}{n}$$

$$\exists c_k \in \mathbb{Q} \in [x_{k-1}, x_k] \Rightarrow \sum_{k=1}^n \underbrace{f(c_k)}_1 \underbrace{(x_{k-1} - x_k)}_{\frac{1}{n}} = 1 \cdot n \cdot \frac{1}{n} = 1.$$

$$\exists c_k \in \mathbb{R} \setminus \mathbb{Q} \in [x_{k-1}, x_k] \Rightarrow \sum_{k=1}^n \underbrace{f(c_k)}_0 \underbrace{(x_{k-1} - x_k)}_{\frac{1}{n}} = 0 \cdot n \cdot \frac{1}{n} = 0.$$

$\Rightarrow \nexists \Rightarrow f$ not i.r.

$$3. a) \int_1^2 \frac{1}{x(x-2)} dx = \lim_{t \nearrow 2} \int_1^t \frac{1}{x(x-2)} dx = \lim_{t \nearrow 2} \left(-\frac{1}{2} \ln t + \frac{1}{2} \ln(t-1) \right)$$

$$= -\frac{1}{2} \ln 2 + \frac{1}{2} \ln 0_+ = -\frac{1}{2} \ln 2 - \infty = -\infty \quad \text{div.}$$

$$\int_1^t \frac{1}{x(x-2)} dx = \frac{1}{2} \int_1^t \frac{x - (x-2)}{x(x-2)} dx = \frac{1}{2} \int_1^t \left(\frac{1}{x-2} - \frac{1}{x} \right) dx = \underline{-\frac{1}{2} \ln t + \frac{1}{2} \ln(t-1)}.$$

$$3. c. \int_0^1 \frac{\ln x}{\sqrt{x}} dx = \lim_{t \searrow 0} \int_t^1 \frac{\ln x}{\sqrt{x}} dx = \dots$$

$$\int_t^1 \frac{\ln x}{\sqrt{x}} dx = 2 \cdot \int_t^1 \frac{\ln x}{2\sqrt{x}} dx = 2 \cdot \int_t^1 (\sqrt{x})' \cdot \ln x dx = 2 \cdot \left(\sqrt{x} \ln x \Big|_t^1 - \int_t^1 \sqrt{x} \cdot \frac{1}{x} dx \right)$$

$$3. d. \int_0^\infty e^{-x} \cdot \sin x dx = \int_0^\infty (e^{-x})' \cdot \sin x dx = \text{parti} \dots$$

$$5.b. \sum_{n=2}^{\infty} \frac{\ln n}{n^2}$$

Integral test.

$$\int \frac{\ln x}{x^2}$$

$$4b. \int_0^{\frac{\pi}{2}} \frac{1}{\cos x} dx$$

conv / div?

$$\cos x = \sin\left(\frac{\pi}{2} - x\right)$$

$$4.c. \int_1^{\infty} \frac{\ln x}{x\sqrt{x^2-1}} dx < \int_1^{\infty} \frac{1}{\sqrt{x^3-x}} dx \quad \text{div.}$$

$$\ln x < \sqrt{x}$$

$$\frac{\ln x}{x\sqrt{x^2-1}} < \frac{\sqrt{x}}{x\sqrt{x^2-1}} = \frac{1}{\sqrt{x}\sqrt{x^2-1}} = \frac{1}{\sqrt{x^3-x}}$$

←

"like" $\int \frac{1}{x^{\frac{3}{2}}}$ conv.

